

CATALOG

Design Capabilities Catalog — Skills, MCP Servers, Knowledge & Uniqueness Approaches

Date: 2026-08-06 · **Purpose:** curated, actionable intelligence on FREE / OSS / no-subscription DESIGN capabilities to incorporate (as git submodules / MCP installs / encoded skills) into a design toolkit **combined with OpenDesign** — to produce unique, enterprise-grade, non-repeatable UI/UX across all platforms, OSes, and app types.

Scope constraint enforced: only genuinely free / open-source / no-subscription items are recommended. Anything paywalled, unlicensed (“all rights reserved” by default), or abandoned (>~12 months no commits) is INCLUDED for the record but **flagged and EXCLUDED from the “incorporate” list**. Every metric (stars, last-push, license) was verified against the live GitHub REST API / repo / npm / docs page during research (2026-08-06). Where a fact could not be confirmed it is marked **unverified**.

How this relates to what already exists here: `_analysis/design-research.md` (2026-08-05) covers *techniques* (scroll-driven animation, view transitions, DTCG tokens, print CSS, responsive images, SEO). This catalog is the complementary *tooling/capability* layer — what to plug in around OpenDesign. `_analysis/OPENDESIGN-ADOPTION-PLAN.md` and the `design-system/` build are the current baseline this augments. Governance context: §11.4.162 mandates OpenDesign as the token source of truth (see companion `INCORPORATION-PLAN.md` for how each item respects that).

Fit rating = 1-5, how strongly the item advances *unique, enterprise-grade, cross-platform* design.

RECOMMENDED TO INCORPORATE — ranked shortlist (value × freeness × ease)

Only truly-free, license-clean, actively-maintained items. Ranked by combined leverage.

#	Item	Type	License	How to add	Why it ranks
1	anthropics/skills (frontend-design, theme-factory, web-artifacts-builder, brand-guidelines, canvas-design, algorithmic-art)	Claude skills	Apache-2.0 (per-skill)	plugin marketplace / copy skill dir / submodule	Official, per mission; from theme-factory aesthetic +
2	material-color-utilities	Color engine (lib)	Apache-2.0	npm dep / submodule in token script	Seed→full a palette + na personalities determinist primitive
3	Chrome DevTools MCP	MCP (QA)	Apache-2.0	claude mcp add (already a plugin here)	Best design a11y/perf + fully local; f proof
4	Playwright MCP	MCP (QA)	Apache-2.0	claude mcp add	Screenshot regression l vendor-back
5	Owl-Listener/ designer-skills	Claude skills (239)	MIT	plugin marketplace (vendor subset)	Full design (research→s modular so complemen
6	Style Dictionary	Token emitter (lib)	Apache-2.0	npm dep / submodule	DTCG→CSS Windows; th fan-out layer OpenDesign
7	color.js (colorjs.io)	Color engine (lib)	MIT	npm dep / submodule	OKLCH-firs APCA/WCA modern ran
8	shadcn MCP (official) + Jpisnice/shadcn-ui-mcp-server	MCP (components)	MIT	npx shadcn@latest mcp / claude mcp add	Registry-dr reuse, mult (React/Vue/
9	GLips/Figma-Context-MCP (Framelink)	MCP (Figma)	MIT	claude mcp add (free Figma PAT)	Only matur Figma-to-co (15.6k★); d

#	Item	Type	License	How to add	Why it ran
10	utopia-core	Fluid scale (lib)	<i>verify LICENSE</i> (sibling MIT)	npm dep in token script	Fluid type/s params; typ personality
11	jezweb/claude-skills (frontend + design-assets)	Claude skills	MIT	plugin marketplace	Tailwind v4 palette/favi marketplac
12	nextlevelbuilder/ui-ux-pro-max-skill	Claude skill	MIT	skill dir / CLI	Widest cross coverage (2 incl. Comp SwiftUI/des
13	chroma.js	Color engine (lib)	BSD-3-Clause	npm dep / submodule	Tiny percept workhorse (color.js)
14	svg.js	Generative SVG (lib)	MIT	npm dep / submodule	Build-time s marks/back project visu
15	draw.io MCP (lgazo) + Mermaid MCP (peng-shawn)	MCP (diagrams)	MIT	npx / claude mcp add	Agent-autho diagram ass
16	x51xxx/coolers-mcp	MCP (color)	MIT	claude mcp add	MD3 + OKL color ops as (low stars, p
17	uxKero/anydesign	Claude skill	MIT	skill dir	Design-to-c extraction f Figma — th counterpart
18	google-fonts-mcp	MCP (fonts)	MIT	claude mcp add -- uvx	Offline font Tailwind typ auth
19	sharp-mcp	MCP (image opt)	MIT	npx	Local AVIF/ (backs the n work in des
20	kenneives/design-token-bridge-mcp	MCP (tokens)	MIT	npm i -g	DTCG→Mat SwiftUI/CS (young; pilo

Knowledge to ENCODE into skills (not installs) — top open-licensed sources: WCAG 2.2 + WAI-ARIA APG (W3C, truly open, legally load-bearing) · Material Design 3 (Apache-2.0/CC-BY-4.0, richest vendorable system) · DTCG token spec (open) · Apple HIG (rules-only, paraphrase — copyrighted) · OKLCH + Radix Colors (MIT) + Open Color (MIT) · Every Layout primitives (concepts) + MDN container queries · Utopia fluid type. Details in Category 3.

CATEGORY 1: Claude Skills / Skill Collections for Design

Star/license/pushed_at verified via GitHub REST API on 2026-08-06.

RECOMMENDED (free + license-clean)

1. anthropics/skills — official design skills — Fit 5/5 - URL: <https://github.com/anthropics/skills/tree/main/skills> - What: Official skills. Design-relevant: frontend-design (distinctive non-templated UI aesthetic), theme-factory (fonts + color themes / design-system theming), web-artifacts-builder (React+Tailwind+shadcn multi-component artifacts), brand-guidelines, canvas-design (visual art→PNG/PDF), algorithmic-art (generative), webapp-testing (Playwright UI verification). - License: **Apache-2.0** per-skill (LICENSE.txt inside frontend-design/ and theme-factory/ confirmed; no root LICENSE so API license field is null). Note: the docx/pdf/pptx/xlsx document skills are “source-available,” NOT OSS — but those are not design skills. - Install: plugin marketplace (.claude-plugin/marketplace.json), or copy a skill folder into ~/.claude/skills/, or git submodule. - Maturity: ~166,638★; last push 2026-07-24; official.

2. Owl-Listener/designer-skills — broadest design-only pack — Fit 5/5 - URL: <https://github.com/Owl-Listener/designer-skills> - What: Largest design-dedicated pack — 239 skills / 88 commands / 33 plugins across design-research, design-systems, ux-strategy, ui-design, interaction-design, prototyping-testing, design-ops, visual-critique. - License: **MIT**. Install: /plugin marketplace add Owl-Listener/designer-skills (also Gemini CLI extensions). - Maturity: 1,980★; last push 2026-06-14; 55 commits.

3. nextlevelbuilder/ui-ux-pro-max-skill — cross-platform design intelligence — Fit 4/5 - URL: <https://github.com/nextlevelbuilder/ui-ux-pro-max-skill> - What: 84 UI styles, 192 palettes, 161 reasoning rules, BM25 style ranking, targeting **22 stacks** (React/Next/Vue/Svelte/SwiftUI/React Native/Flutter/Tailwind/shadcn/**Jetpack Compose**/Angular/JavaFX/WPF/WinUI3/Avalonia/Uno...). - License: **MIT** (optional PayPal donations; free ui-ux-pro-max-cli on npm; no subscription gate). - Maturity: ~114,056★ (API-reported; single-maintainer); last push 2026-08-06. - Caveat: single-maintainer, donation-driven; a large “styles catalog” can drift generic unless paired with OpenDesign tokens.

4. jezweb/claude-skills — Tailwind v4 / shadcn / design-assets — Fit 4/5 - URL: <https://github.com/jezweb/claude-skills> - What: 52 skills / 11 plugins. Design-relevant: frontend (Tailwind v4

theming, shadcn, landing pages, React 19, visual design-quality review, RN+Expo) and design-assets (color-palette gen, favicon, SVG icon sets, image processing/gen). - License: **MIT**. Install: /plugin marketplace add jezweb/claude-skills → /plugin install frontend@jezweb-skills. - Maturity: 957★; last push 2026-07-02; 717 commits. Sibling: secondsky/claude-skills (MIT, 204★, pushed 2026-08-06).

5. uxKero/anydesign — design-to-code / token extraction — Fit 4/5 - URL: <https://github.com/uxKero/anydesign> - What: Analyzes any image, website URL, or Figma file → structured design.md with design system + tokens + component inventory. - License: **MIT**. Install: copy skill folder into ~/.claude/skills/. - Maturity: 151★; last push 2026-06-11.

6. mattbx/shadcn-skills — shadcn discovery & review — Fit 3/5 - URL: <https://github.com/mattbx/shadcn-skills> · **MIT** · 15★ · last push 2026-04-27. Discover 1,500+ shadcn components before building custom; review against shadcn patterns/themes. Narrow but useful for reuse/consistency.

7. wholiver/swiftui-design-skill — SwiftUI anti-slop — Fit 3/5 - URL: <https://github.com/wholiver/swiftui-design-skill> · **MIT** · 164★ · last push 2026-05-01. Apple-platform front-end design with “six rules against AI sloppiness.” Single-platform scope.

8. rohitg00/awesome-claude-design — aesthetic DESIGN.md packs — Fit 3/5 - URL: <https://github.com/rohitg00/awesome-claude-design> · **MIT** · 963★ · last push 2026-04-23. 28 DESIGN.md exemplars by aesthetic family + “anti-slop” fingerprint toolkit. Prompt scaffolding, not installable skills — good for aesthetic direction / avoiding generic output.

FLAGGED / EXCLUDED (license or content issues)

- **tommyjepson/awesome-ux-skills** — 30 UX/product frameworks (personas, journey maps, double diamond, heuristic eval, Dieter Rams, AI-product patterns). 131★, pushed 2026-07-28. **NO LICENSE file → all-rights-reserved; unsafe to vendor.** Excellent content — clarify license before use. Fit 3/5 but blocked.
- **Awesome-list indexes** (discovery only, licenses unconfirmed): ComposioHQ/awesome-claude-skills (71,942★, README says Apache-2.0 but API reports none), travisvn/awesome-claude-skills (14,536★, no license), wilwaldon/Claude-Code-Frontend-Design-Toolkit (604★, no license).
- **obra/superpowers** (267,850★, MIT, very active) — **NOT a design pack**: skills are dev methodology (TDD, debugging, worktrees). Adopt only for workflow discipline. The old obra/superpowers-skills is **ARCHIVED** (2025-10-14) — do not depend on it.

- **Figma official Claude plugin** (figma:figma-generate-design, etc.) — vendor-distributed, requires Figma account/auth; no verifiable OSS license/public repo. Treat as proprietary integration (present in this environment as a plugin).
 - Vendor-scoped skills (not general): Qt (qt-figma-*, qt-ui-design), Atlassian forge:atlassian-design-tokens, Wix wix-design-system, Shopify Polaris — useful only within their own SDKs.
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CATEGORY 2: Free Design MCP Servers

“Truly free” = works with no paid subscription. Paid/hosted-gated servers are flagged and excluded.

2A — Figma / Icons / Color / Tokens / Image-gen

Figma access

- **GLips/Figma-Context-MCP (Framelink)** — Fit 5/5 — <https://github.com/GLips/Figma-Context-MCP> — **MIT** — reads Figma layout/design data via REST → simplified JSON for agents (get_figma_data, download_figma_images). **TRULY FREE: free Figma Personal Access Token, any plan** (no Dev Mode seat). Install: `claude mcp add framelink-figma -- npx -y figma-developer-mcp --figma-api-key=YOUR-KEY --stdio`. 15,597★; last push 2026-07-03. The de-facto free design-to-code bridge.
- **open-figma-mcp (Shivam990q)** — Fit 2/5 — <https://github.com/Shivam990q/open-figma-mcp> — MIT — free PAT; smaller-token alt + token extraction. Only 3★, single author — unproven vs Framelink.
- **EXCLUDED — Official Figma Dev Mode MCP** — <https://developers.figma.com/docs/figma-mcp-server/> — proprietary. Free-tier ~6 calls/month (unusable); real use needs a **paid Dev/Full seat**, and Figma states it “will eventually be usage-based paid.” Fails the no-subscription bar.

Icons

- **imjac0b/iconify-mcp-server** — Fit 3/5 — <https://github.com/imjac0b/iconify-mcp-server> — **GPL-3.0** — searches Iconify’s 200k+ icons; free, no key. 13★; last push 2025-11-22. Cleanest single-source with a real license (GPL = fine as external tool, note if vendoring code).
- **awssat/mcp-universal-icons** — <https://github.com/awssat/mcp-universal-icons> — **NO LICENSE** (all-rights-reserved) — best breadth (~10 sets: Material Symbols, Lucide, Tabler, Heroicons, Phosphor, FontAwesome, Feather, Bootstrap, Remix, Iconoir) + Tailwind injection; free via Iconify API. 5★; last push 2026-01-25. Blocked by missing license.

- **Osmansiddiquer/iconify-mcp** (README says MIT but no LICENSE file; 0★) and **agentic-ph/icon-mcp** (MIT but last push 2025-08-01 → stale/abandoned) — both weak.

Color / palette

- **x51xxx/coolors-mcp** — Fit 4/5 — <https://github.com/x51xxx/coolors-mcp> — **MIT** — MD3 support, HCT/OKLCH perceptual color, Delta-E, CSS theme matching, image color extraction, WCAG. **Fully local, no key.** Install: `claude mcp add coolors -- npx -y @trishchuk/coolors-mcp`. 2★; last push 2026-05-20. Most design-system-relevant color server; low stars → pilot first.
- **rog0x/mcp-color-tools** — Fit 3/5 — <https://github.com/rog0x/mcp-color-tools> — **MIT** — HEX/RGB/HSL/CMYK convert, palette gen (complementary/analogous/triadic/split), WCAG contrast, gradients. Local, no key. 0★; last push 2026-03-21.
- **ColorMcp (keyurgolani)** — MIT, 0★, README “TBD” — verify it works. **color-scheme-mcp (deepakkumardewani)** — NO LICENSE + last push 2025-05-31 (abandoned).

Design tokens

- **kenneives/design-token-bridge-mcp** — Fit 4/5 — <https://github.com/kenneives/design-token-bridge-mcp> — **MIT** — ingests Tailwind/CSS/Figma/**W3C DTCG** → emits Material 3 (Kotlin), SwiftUI, Tailwind, CSS Variables. Local, no key. Install: `npm install -g design-token-bridge-mcp`. 5★; last push 2026-03-25. Only real *MCP server* for DTCG cross-platform output; young → pilot.
- Note: **Style Dictionary** and **tokens-studio/sd-transforms** are libraries/CLI, **not MCP servers** (see Category 4 / knowledge).

Image / asset generation (free / local)

- **joenorton/comfyui-mcp-server** — Fit 4/5 — <https://github.com/joenorton/comfyui-mcp-server> — **Apache-2.0** — drives a **local ComfyUI** (Stable Diffusion) for text-to-image. **Fully local, no paid API** (needs your own ComfyUI + models/GPU). 396★; last push 2026-02-17.
- **artokun/comfyui-mcp** — Fit 4/5 — <https://github.com/artokun/comfyui-mcp> — **MIT** — local-first ComfyUI control plane (image/video/audio, authors+runs workflows, 178 tools). 508★; last push 2026-08-06 (very active). Heavier than joenorton’s.

2B — Diagramming / Screenshot / Fonts / Illustration / Motion / Components / Layout / Image-opt

Diagramming / whiteboard

- **draw.io MCP (lgazo)** — Fit 4/5 — <https://github.com/lgazo/drawio-mcp-server> — **MIT** — CRUD draw.io/diagrams.net elements; built-in editor or browser-extension bridge. Fully local, no key. Install: `npx -y drawio-mcp-server --editor`. ~1.4k★, ~266 commits (most-adopted community draw.io MCP).
- **Mermaid MCP (peng-shawn)** — Fit 4/5 — <https://github.com/peng-shawn/mermaid-mcp-server> — **MIT** — Mermaid code → PNG/SVG via headless Puppeteer, fully local. Install: `claude mcp add mermaid -- npx -y @peng-shawn/mermaid-mcp-server`. ~232★.
- **Excalidraw MCP (official)** — Fit 4/5 — <https://github.com/excalidraw/excalidraw-mcp> — **MIT** — streams editable hand-drawn diagrams with live canvas control. Fully local (optional hosted endpoint not required). ~5.1k★; build step (pnpm) required.

Screenshot / visual-diff / browser QA

- **Chrome DevTools MCP** — Fit 5/5 — <https://github.com/ChromeDevTools/chrome-devtools-mcp> — **Apache-2.0** — live Chrome automation, DOM/console/network, **Lighthouse audits**, perf traces, screenshots. Fully local (perf-trace URLs go to Google CrUX API + anon usage stats — both opt-out). Install: `claude mcp add chrome-devtools --scope user npx chrome-devtools-mcp@latest`. ~48.6k★. Already a plugin in this environment.
- **Playwright MCP** — Fit 5/5 — <https://github.com/microsoft/playwright-mcp> — **Apache-2.0** — accessibility-tree-driven browser automation + screenshots. Fully local. Install: `claude mcp add playwright npx @playwright/mcp@latest`. ~35.9k★.
- **MCP Image Compare (leky90)** — Fit 3/5 — <https://github.com/leky90/mcp-image-compare-server> — **MIT** — Pixelmatch pixel-diff of two images or URL-screenshot-vs-baseline. Local (needs playwright install chromium). ~3★ (immature — Playwright's own `toHaveScreenshot` is more battle-tested).

Fonts

- **google-fonts-mcp (sliday)** — Fit 4/5 — <https://github.com/sliday/google-fonts-skill> (PyPI `google-fonts-mcp`) — **MIT** — searches ~1,923 Google Fonts, suggests pairs, emits CSS/Tailwind type systems. Fully local/offline, no key. Install:

claude mcp add google-fonts -- uvx google-fonts-mcp. ~9★
(niche but self-contained).

Illustration / SVG

- **svgo-mcp (undirectlookable)** — Fit 2/5 — <https://github.com/undirectlookable/svg-mcp> — MIT — SVGO optimize/minify. 0★. Trivially replaceable by a svgo Bash step — low value as a server.
- **svgr-mcp (pchalupa)** — Fit 2/5 — <https://github.com/pchalupa/svgr-mcp> — license **unverified** — SVG→React via SVGR. Niche.

Motion / Lottie

- **mcp-server-lottiefiles (junmer)** — Fit 2/5 — <https://github.com/junmer/mcp-server-lottiefiles> — MIT — searches LottieFiles. **Almost certainly remote-dependent; whether a token is required is unverified.** ~19★. The vendor “Lottie Creator MCP” is a hosted product tied to a LottieFiles account — **excluded** (not free/local). Prefer CSS/SVG motion (per design-research.md).

UI component libraries

- **shadcn MCP (official)** — Fit 5/5 — <https://ui.shadcn.com/docs/mcp> — shadcn/ui **MIT** — browse/search/install components by NL. Fully local & free for public registries (keys only for private). Install: `npx shadcn@latest mcp init --client claude`.
- **Jpisnice/shadcn-ui-mcp-server** — Fit 5/5 — <https://github.com/Jpisnice/shadcn-ui-mcp-server> — **MIT** — component source/blocks/demos/metadata across **React, Svelte, Vue, React Native**. Local; optional GitHub token only to raise rate limits. Install: `claude mcp add shadcn -- npx -y @jpisnice/shadcn-ui-mcp-server`. ~2.9k★.
- **Magic UI MCP (official)** — Fit 4/5 — <https://github.com/magicuidesign/mcp> — **MIT** — search/install Magic UI animated/marketing components. Local, no key. `npx @magicuidesign/mcp@latest`. ~200★.
- **MUI MCP (official)** — Fit 3/5 — <https://mui.com/material-ui/getting-started/mcp/> — MUI core MIT (MCP pkg license unstated — verify) — serves MUI docs/examples (docs-grounding, not codegen). Local. `npx -y @mui/mcp@latest`.
- **EXCLUDED — 21st.dev Magic MCP** — <https://github.com/21st-dev/magic-mcp> — **license unspecified** — 10,000+ React/Tailwind components + AI UI gen, but **requires a 21st.dev API key** and README references “paid code retrieval”; free-tier limits undocumented. Fails the no-subscription bar.

Layout / grid — no mature free MCP exists. - **CarbonoDev/tailwindcss-mcp-server** — Fit 3/5 — <https://github.com/CarbonoDev/tailwindcss-mcp-server> — MIT, ~38★, local — Tailwind utility/grid/flex guidance + class optimization. Most credible of a thin field. Others pull in paid Gemini keys or are abandoned. Treat layout as a capability of the component/Tailwind servers.

Image optimization

- **sharp-mcp (greatSumini)** — Fit 3/5 — <https://github.com/greatSumini/sharp-mcp> — **MIT** — libvips/Sharp resize/convert/metadata/color-extract + optional local ML bg-removal. Fully local, no key. `npx -y sharp-mcp`. ~11★ (lib is industry-standard).
 - **mcp-image-optimizer (piephai)** — <https://github.com/piephai/mcp-image-optimizer> — license **unverified** — Sharp optimize/resize/convert + LQIP placeholders. Comparable; verify license. (Squoosh forks exist but Squoosh upstream is effectively unmaintained → archival risk.)
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CATEGORY 3: Design Knowledge / Method Resources (to ENCODE into skills)

Distinguishes **truly open** (CC/MIT/Apache — vendorable/quotable) from **free-to-read but copyrighted** (readable; distill rules in your own words) from **PAID**. Facts/heuristics aren't copyrightable — encode the *rules*, not the prose. Vendor sites are usually copyrighted while their sibling code repos are MIT/Apache.

3.1 All-platform UI/UX guidelines

- **Apple HIG** — <https://developer.apple.com/design/human-interface-guidelines> — iOS/iPadOS/macOS/watchOS/tvOS/visionOS. **Free-to-read, COPYRIGHTED** (paraphrase rules only). Fit 5.
- **Material Design 3** — <https://m3.material.io/> — color roles, type scale, motion/easing, components, tokens, adaptive layout. **TRULY OPEN — Apache-2.0 / CC-BY-4.0 (vendorable w/ attribution)**. Fit 5.
- **Microsoft Fluent 2** — <https://fluent2.microsoft.design/> (site copyrighted) + **microsoft/fluentui** code **MIT** (<https://github.com/microsoft/fluentui>). Fit 4.
- **Windows app design** — <https://learn.microsoft.com/en-us/windows/apps/design/> — MS Learn docs generally CC-BY-4.0 (confirm per page). Fit 3.

- **GNOME HIG** — <https://developer.gnome.org/hig/> — Linux/GTK conventions; GNOME docs typically CC-BY-SA. Fit 3.
- **WCAG 2.2** — <https://www.w3.org/TR/WCAG22/> — accessibility standard (W3C Rec, 2024-12-12). **Free + W3C Document License** (freely readable/redistributable; normative text unmodifiable). Legal baseline. Fit 5.
- **WAI-ARIA APG** — <https://www.w3.org/WAI/ARIA/apg/> — accessible design *patterns* (keyboard models + ARIA for menus/dialogs/combobox/tabs...). **TRULY OPEN**, code examples vendorable. Fit 5.
- **WHATWG HTML** — <https://html.spec.whatwg.org/multipage/> — **CC-BY-4.0**. Fit 3. · **MDN** — <https://developer.mozilla.org/> — content CC-BY-SA, code CC0. Fit 4.

3.2 Design-system methodologies

- **Atomic Design (Brad Frost)** — <https://atomicdesign.bradfrost.com/> — atoms→pages taxonomy. **Free-to-read, COPYRIGHTED** (distill taxonomy). Fit 5.
- **DTCG token spec** — <https://www.designtokens.org/> (drafts <https://www.designtokens.org/TR/>) — standard token JSON; first stable **2025.10. Open** (W3C Community Group). Fit 5.
- **Style Dictionary** — <https://styledictionary.com/> — DTCG→CSS/iOS/Android/JS. **TRULY OPEN — Apache-2.0**. Fit 5.
- **IBM Carbon** — <https://carbondesignsystem.com/> — enterprise system (tokens, components, a11y, data-viz). **Apache-2.0**. Fit 5.
- **Shopify Polaris** — <https://shopify.dev/docs/api/polaris> — code **MIT**; docs free. Fit 4. · **GitHub Primer** — <https://primer.style/> — code **MIT**. Fit 4.
- **Every Layout** — <https://every-layout.dev/> — **PAID (\$69)**; intro chapters free (concepts encodable). Fit 5. · **Refactoring UI** — <https://www.refactoringui.com/> — **PAID (\$99-149)**; buy as team reference, never copy. Fit 4.

3.3 Motion / animation

- **Material 3 Motion** — <https://m3.material.io/styles/motion/overview> — easing tokens, durations, choreography. **Open (Apache-2.0/CC-BY-4.0)**. Fit 5.
- **easings.net** — <https://easings.net/> — 31 easings + cubic-bezier/CSS. **MIT** (<https://github.com/ai/easings.net>) — vendorable table. Fit 4.
- **prefers-reduced-motion (MDN)** — <https://developer.mozilla.org/en-US/docs/Web/CSS/@media/prefers-reduced-motion> — reduced-motion + WCAG 2.3.3 gate. **CC-BY-SA**. Fit 5.
- **Disney 12 principles** — Wikipedia (CC-BY-SA) summary; original book is PAID. Fit 4. (Val Head's book/articles are paid/

copyrighted — use Material Motion + easings.net + WCAG instead.)

3.4 Color

- **OKLCH picker (Evil Martians)** — <https://oklch.com/> — OKLCH/LCH perceptual space + P3 fallbacks. Free/OSS. Encode OKLCH as the palette model. Fit 5.
- **WCAG 2.2 contrast (1.4.3/1.4.11)** — the **legally authoritative** ratios (4.5:1 text, 3:1 large/UI). Enforce this. Fit 5.
- **APCA** — <https://github.com/Myndex/SAPC-APCA> — perceptual Lc contrast; **bespoke beta-license, pulled from WCAG 3 draft (2023), undetermined as of 2026**. Advisory only, not authoritative. Fit 4.
- **Material 3 color roles** — <https://m3.material.io/styles/color/system/overview> — semantic roles + tonal palettes. **Open**. Fit 5.
- **Radix Colors** — <https://www.radix-ui.com/colors> — 12-step accessible scales, APCA-tuned, auto dark, P3. Code **MIT**. Fit 5.
- **Open Color** — <https://yeun.github.io/open-color/> — 13 hues × 10 shades. **MIT** (v1.9.1). Fit 4.

3.5 Typography

- **Utopia** — <https://utopia.fyi/> — fluid `clamp()` type/space scales; free calculators (concepts encodable). Fit 5.
- **web.dev Variable Fonts** — <https://web.dev/articles/variable-fonts> — **content CC-BY-4.0, code Apache-2.0**. Fit 4.
- **Material 3 Typography** — <https://m3.material.io/styles/typography/overview> — type-scale roles. **Open**. Fit 4. · **MDN CSS text/font** — CC-BY-SA. Fit 3.

3.6 Layout

- **Every Layout primitives** (Stack/Cluster/Sidebar/Switcher/Cover/Grid) — <https://every-layout.dev/> — PAID but *concepts* encodable. Fit 5.
- **MDN Container Queries** — https://developer.mozilla.org/en-US/docs/Web/CSS/CSS_containment/Container_queries — `container-type/@container/cqi`. **CC-BY-SA**. Component-level responsiveness. Fit 5.
- **Material 3 Layout & spacing** — <https://m3.material.io/foundations/layout/understanding-layout/overview> — authoritative free source of the **4/8dp grid**. **Open**. Fit 5. · **MDN Grid/Flexbox** — CC-BY-SA. Fit 4.

Highest-leverage to encode (open × cross-platform value):

- (1) WCAG 2.2 + WAI-ARIA APG — hard a11y gate + widget specs;
- (2) Material Design 3 whole system — richest vendorable defaults

(color roles, type scale, motion, 8pt grid); (3) DTCG + Style Dictionary — token interoperability backbone; (4) Apple HIG rules-only — platform-adaptation checklist for cross-platform nativeness; (5) OKLCH + Radix + Open Color + WCAG contrast — color stack; (6) Every Layout primitives + container queries + 8pt grid — layout engine; (7) Utopia + variable-fonts — fluid type.

CATEGORY 4: Approaches for Unique, Non-Repeatable Per-Project Design

The organizing idea: reduce a project's aesthetic to a small **parameter vector** / “**design DNA**” **seed**, deterministically expand it into a full token set, and add build-time seeded generative accents — so every project is distinct-but-on-brand and reproducible (same seed = same design).

(A) Parametric token / color tools (seed → full token set)

- **material-color-utilities** — Fit 5 — <https://github.com/material-foundation/material-color-utilities> — **Apache-2.0** — Google's M3 dynamic color: one seed → HCT space → tonal palettes + role-mapped light/dark scheme. Ships named **scheme variants** (Vibrant/Expressive/Fidelity/Neutral/Monochrome/Fruit-Salad) — each a per-project “personality” dial. TS/Java/Dart/C++/Swift/Kotlin. **The strongest deterministic uniqueness primitive.**
- **color.js (colorjs.io)** — Fit 5 — <https://github.com/color-js/color.js> — **MIT** — OKLCH-first conversion/interpolation, gamut-safe P3, APCA/WCAG contrast (by CSS Color spec editors). The future-proof ramp/harmony engine.
- **chroma.js** — Fit 5 — <https://github.com/gka/chroma.js> — **BSD-3-Clause** — tiny perceptual `chroma.scale()` (Lab/Lch), Bezier interpolation, correctLightness. Workhorse for custom harmony ramps (alt/base to color.js).
- **Adobe Leonardo** (@adobe/leonardo-contrast-colors) — Fit 4 — <https://github.com/adobe/leonardo> — **Apache-2.0 (open source, NOT the paid Adobe CC product)** — generates ramps from *target contrast ratios*; adaptive. Personality = contrast curve + background luminance. Optional add for contrast-driven theme modes.
- **Radix Colors custom generator** — Fit 4 — <https://github.com/radix-ui/colors> (tool <https://www.radix-ui.com/colors/custom>) — **MIT** — role-semantic 12-step scales from 1-2 seeds; the “safe/restrained” personality.
- **utopia-core** — Fit 5 (technique) — <https://github.com/trys/utopia-core> — license **to confirm on repo** (npm 403; sibling `tailwind-utopia` is MIT) — `calculateTypeScale/SpaceScale/Clamp`

→ fluid `clamp()` from params. Type-scale ratio (1.2 vs 1.333 vs 1.5) + spacing multiplier = a strong personality axis. **Verify LICENSE before submoduling.**

- **Style Dictionary** — Fit 5 — <https://styledictionary.com/> / <https://github.com/amzn/style-dictionary> — **Apache-2.0** — the emitter (not a generator): one generated token set → every platform + per-project overrides. Sits on top of OpenDesign output.
- **Tailwind palette generators** — Fit 3 — tints.dev (MIT, <https://github.com/SimeonGriggs/tints.dev>, has JSON API), uicolors.app, [tailwindcss-palette-generator](https://tailwindcss-palette-generator.com), tailwind.ink — one hex → 11-step scale. Only for Tailwind-native projects; less rigorous than MCU/OKLCH.

(B) Theme-variation strategies (parameters → many distinct themes)

- **Seed + harmony rules** (Fit 5): hue rotation — complementary $+180^\circ$, analogous $\pm 30^\circ$, triadic $\pm 120^\circ$, split-comp. MCU does this internally; or roll your own with [color.js](https://color.js.org)/chroma.js. The *harmony rule chosen* is a per-project lever.
- **Scheme “variant” presets** (Fit 5): MCU’s Vibrant/Expressive/Neutral/Monochrome/Fidelity — cheapest reliable “make it feel different” switch.
- **Type-scale ratio + font pairing** (Fit 5): ratio sets hierarchy drama; the pairing (geometric-sans display + humanist body vs serif display + grotesque body vs mono accents) sets voice. Curate an OSS font-pair matrix (Google Fonts) keyed to brand adjectives — **highest visual differentiation per unit effort**. Refs: <https://madegooddesigns.com/font-pairing/> , <https://www.axiswebart.com/blog/fonts-can-express-personality-traits/>
- **Shape/roundness + spacing density** (Fit 5): radius base (0 sharp/technical → 8-12px friendly → pill playful) + spacing multiplier (compact vs airy) — big feel change, near-zero risk. Frames these as *ranges*: <https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/11853653> (style-grammars patent).
- **Contrast/mode variants** (Fit 4): Leonardo contrast curve → “premium dark high-contrast” vs “airy light.”
- **Design-personality parameter model** (Fit 4, framework): adjectives (modern/trustworthy/playful/editorial/technical) → {hue temp, chroma, radius, scale ratio, font class, density}. Methodology, not a tool — design your own schema.

(C) Generative aesthetics libs (build-time marks / backgrounds / textures)

- **svg.js** — Fit 5 — <https://github.com/svgdotjs/svg.js> — MIT — clean resolution-independent SVG creation; seed a

deterministic RNG → per-project SVG texture/mark/divider, colors from tokens. Best for *production*.

- **p5.js** — Fit 3 — <https://github.com/processing/p5.js> — **LGPL-2.1** — creative coding; use at *build time* for offline SVG/PNG assets (note copyleft if ever bundled into shipped JS).
- **seeded-noise** — Fit 3 — <https://github.com/andyinabox/seeded-noise> — seeded simplex noise for reproducible organic textures (confirm license). · **ribbit** — Fit 3 — <https://ribbit.sn4pe.com> — **MIT** — hash string→seed→identical SVG mark (deterministic avatars/marks).
- Reference: georgedoescode “Generative SVG Starter Kit” — <https://dev.to/georgedoescode/a-generative-svg-starter-kit-5cm1> (technique).
- **Caveat:** keep generative art at *build time* producing static SVG assets — protects performance and accessibility (consistent with design-research.md GPU-only / no-jank rules).

(D) Seed / “design-DNA” concepts

- **Material 3 dynamic color IS the canonical shipping example** of “one seed → whole accessible system,” OSS Apache-2.0 — proof the approach is enterprise-grade, not a blog fantasy.
- **Deterministic-seed principle** (verified via ribbit / seeded-noise): same seed + options + version ⇒ identical output. Derive a project seed = hash(project-name + domain) → color seed, hue offsets, scale ratio, radius, pattern params. Reruns stable; projects diverge.
- **No single OSS “brand-in-a-box”** does color+type+space+shape+marks end-to-end — the pieces exist (MCU + Utopia + Style Dictionary + svg.js); assembly is our work. “Design tokens from a seed for AI agents” write-ups (<https://www.mindstudio.ai/blog/design-token-system-ai-agents-brand-visuals> , <https://www.contentful.com/blog/design-token-system/>) are methodology, not code.
- **LLM/agent-driven variation** (credible research, not a product): **CPT — “Controllable & Editable Design Variations with Language Models”** <https://arxiv.org/html/2604.04380> — choose which attributes vary (color/font/layout) under brand constraints. Practitioner guidance: <https://www.uxpin.com/studio/blog/connect-design-system-llms-on-brand-ui/> . **Takeaway:** don’t free-form “invent a design”; give the agent a **bounded parameter space** and have it select + justify values → deterministic, on-brand, unique.

Proposed recipe — “make every Vasic project look distinct on top of OpenDesign”

See INCORPORATION-PLAN.md §“Parametric uniqueness layer” for the full pipeline. In brief: a container-only Node token-generator takes a per-project **design-DNA vector** {seedHue, mcuVariant, typeRatio, spaceMultiplier, radiusBase, fontPairId, contrastMode} (deterministically derived from a seed) → **material-color-utilities** for HCT palettes + variant personality → **color.js/chroma.js** for custom OKLCH harmony ramps + P3 → **utopia-core** for fluid type/space → radius+density tokens → **Style Dictionary** emits the CSS custom properties the sites consume, with **OpenDesign remaining the structural/semantic source of truth** (this layer only supplies values, honoring §11.4.162 via upstream emission — see plan). Plus **svg.js** build-time seeded marks/backgrounds for a visual signature. The parameter space + adjective→range mapping is encoded as a **theming-designer skill** (constrained selection, not free-form).

Consolidated honesty flags (do NOT incorporate as-is)

- **Paywalled / hosted-gated (excluded):** Official Figma Dev Mode MCP (needs paid seat), 21st.dev Magic MCP (API key + paid retrieval), LottieFiles Creator MCP (hosted account), Refactoring UI (\$99–149), Every Layout (\$69 — concepts only), *Disney Illusion of Life* / Val Head books.
- **No license = all-rights-reserved (unsafe to vendor until clarified):** awssat/mcp-universal-icons, Osmansiddiquier/iconify-mcp, deepakkumardewani/color-scheme-mcp, tommyjepsen/awesome-ux-skills, most awesome-list indexes (Composio/travisvn/wilwaldon).
- **Copyrighted-but-free-to-read (encode rules, don’t copy text):** Apple HIG, Fluent 2 site, Atomic Design online book.
- **Advisory/unsettled:** APCA (bespoke beta license, pulled from WCAG 3 draft) — use as supplementary; enforce WCAG 2.2 ratios.
- **Stale / abandoned (>~12mo):** agentic-ph/icon-mcp (2025-08), color-scheme-mcp (2025-05), obra/superpowers-skills (ARCHIVED 2025-10), Squoosh upstream.
- **Immature (≤5★, single author — pilot before standardizing):** colors-mcp, mcp-color-tools, design-token-bridge-mcp, open-figma-mcp, google-fonts-mcp, sharp-mcp, MCP Image Compare, svgo-mcp.
- **Unverified licenses to confirm before vendoring:** utopia-core, @mui/mcp package, svgr-mcp, piephai/mcp-image-optimizer, seeded-noise, tailwindcss-palette-generator.

- **Not what the name implies:** obra/superpowers (no design skills); Style Dictionary / sd-transforms / DTCG (libraries/specs, not MCP servers); layout/grid MCPs (no mature free option — use component/Tailwind servers).

2026-08-07 EXPANSION — additional shortlist (license-tagged)

A second research pass broadened capability coverage (icons, motion, print, TUI, XR, data-viz, full design systems) and surfaced more FREE/no-key MCPs. Everything below is **truly free / OSS / no subscription** unless a caveat is stated. This section is **additive** to the shortlist above.

New generator / library deps (added to generators/package.json)

Item	License	Role
Open Props	MIT	Ready-made design-token breadth (color/space/type/easing/shadow/gradient) complementing the seeded generator.
culori	MIT	Fast, tree-shakeable color-space math (OKLCH/OKLAB/P3/Delta-E) alongside colorjs.io/chroma.js.
Motion (motion.dev, formerly Framer Motion)	MIT	The OSS motion engine we standardize on — replaces non-OSI GSAP .
@iconify/json + 9-set icon breadth	MIT wrapper (each set keeps its own permissive license)	Offline icon-data engine. Concrete permissive sets pinned: lucide (ISC) , @tabler/icons (MIT) ,

Item	License	Role
Paged.js (pagedjs)	MIT	@phosphor-icons/core (MIT) (+ Material Symbols, Heroicons, Feather, Bootstrap, Remix, Iconoir via Iconify data). Print / paged-media (CSS Paged Media polyfill) for PDF/print output.

EXCLUDED from deps: GSAP (non-OSI “no-charge” license with redistribution/competitor restrictions — not OSS; Motion replaces it), **tints.dev** (fine hosted tool, but prefer local culori/chroma ramps), **Animate.css** (MIT but a prebuilt grab-bag that undercuts the seeded/Motion approach), **Vivliostyle** (**AGPL-3.0** copyleft — Paged.js is the MIT print path).

New **FREE** MCPs (added to mcp/INSTALL.md §3b)

MCP	License	Role
better-icons	MIT	200k+ icon search/insertion (Iconify-backed) — iconographer.
mcp-excalidraw-server	MIT	Local editable wireframes/diagrams — ux-flow-designer.
mcp-image-optimizer	MIT	Local image optimize/resize/convert (Sharp superset) — iconographer.
hig-mcp	MIT	Apple HIG search + audit (the FREE replacement for the paid Apple-UI MCP) — design-qa-auditor.

MCP	License	Role
wcag-guidelines-mcp	MIT on npm, NO LICENSE file (CAVEAT — external use only)	WCAG 2.2 success-criteria lookup — design-qa-auditor.
design-systems (southleft, hosted HTTP)	MIT	Design-systems patterns/components/tokens knowledge — layout-architect/theming-designer.
tailwindcss-mcp-server	MIT	Tailwind utility/grid/flex guidance — layout-architect.
@just-every/mcp-screenshot-website-fast	MIT	Fast full-page screenshots — design-qa-auditor.
@antv/mcp-server-chart	MIT (hosted-render default; set VIS_REQUEST_SERVER for offline)	25+ chart types — ux-flow-designer/design-qa-auditor.
daisyui-mcp, registry-directory-mcp, joenorton/comfyui-mcp-server	MIT / MIT / Apache-2.0	Submodule/build-from-source candidates; ComfyUI is local-only (needs your own GPU).

New reference / full design-system repos (submodule candidates, submodules/PLAN.md §C-bis)

- **Full systems:** Carbon (**Apache-2.0**), Fluent UI (**MIT**), PatternFly (**MIT**), USWDS (**public domain**), GOV.UK Frontend (**MIT**).
- **Headless quartet:** Radix Primitives (**MIT**), Ark UI (**MIT**), Base UI (**MIT**), Headless UI (**MIT**).
- **TUI:** Charm lipgloss (**MIT**), bubbletea (**MIT**), Textual (**MIT**).
- **Print:** Paged.js (**MIT**).
- **Data-viz:** D3 (**ISC**), Observable Plot (**ISC**), Vega/Vega-Lite (**BSD-3-Clause**), ECharts (**Apache-2.0**), visx (**MIT**).
- **XR:** A-Frame (**MIT**), react-three-fiber (**MIT**), MRTK (**MIT**).

- **npm-dep-preferred default:** pin as npm deps where a package exists; submodule only for source-level reference/patching.

EXCLUDED (2026-08-07 pass) — paid / no-license / copyleft / abandoned

- **Paid / subscription:** Apple UI “design” MCP (paid → use hig-mcp), Figma Dev Mode MCP (paid seat), 21st.dev Magic MCP (API key + paid retrieval), paid cloud image APIs (TinyPNG / hosted bg-removal / cloud image-gen).
- **Copyleft (kept out of vendored tree):** **storybook-mcp-server (AGPL-3.0)**, GPL iconify servers (imjac0b et al.), **Vivliostyle (AGPL-3.0)**, GSAP (non-OSI).
- **No LICENSE file (all-rights-reserved / caveat):** awssat/mcp-universal-icons, Osmansiddiquier/iconify-mcp, deepakkumardewani/color-scheme-mcp; **wcag-guidelines-mcp** and **material3-style servers** declare MIT on npm but ship no LICENSE file → external process only, do not vendor source.
- **Abandoned / archived:** agentic-ph/icon-mcp, archived puppeteer MCP, Squoosh-based optimizers.
- **Hosted-render caveat:** @antv/mcp-server-chart renders remotely unless VIS_REQUEST_SERVER is set; local image-generation (ComfyUI) needs your own GPU.